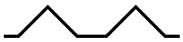
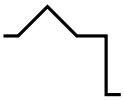
Math Kangaroo Practice

Grade 3-4 · 24 questions · harder · seed 37503

Part A: Each question is worth 3 points

1. Lena cut a strip of paper along this line.

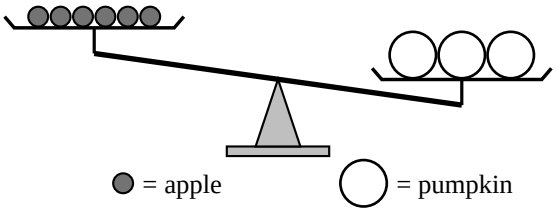
Which piece **cannot** come from the cut line?

A. ***B.***  ***C.*** ***D.***  ***E.***

2. Omar, Mia, Lucas, Julien and Leo each have a different pet: a cat, a goldfish, a dog, a hamster and a turtle. Omar has a pet with no legs at all. Leo has a pet that barks. Julien doesn't like hamsters. Mia has a furry animal. Julien has a furry animal.
Who has a cat?

A. Omar ***B.*** Julien ***C.*** Leo
D. Lucas ***E.*** Impossible to determine

3. On one plate of a balance there are 6 apples and on the other there are 3 pumpkins. When we put 2 more pumpkins, exactly like the others, on the plate with the apples, the balance is level.



One pumpkin weighs the same as how many apples?

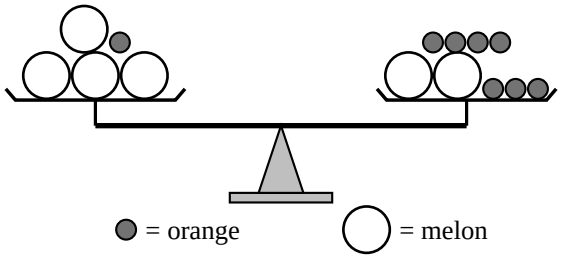
A. 12 apples ***B.*** 6 apples ***C.*** 4 apples ***D.*** 3 apples ***E.*** 2 apples

4. Emma, Julien, Tom, Mia and Noah were born in different months: July, January, April, December and March. Noah and Tom both were born in spring. Mia has a birthday in summer. Noah was not born in April. Julien and Mia both were born in a month that starts with J.
Who was born in December?

A. Tom ***B.*** Julien ***C.*** Mia
D. Emma ***E.*** Impossible to determine

Part B: Each question is worth 4 points

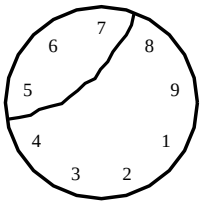
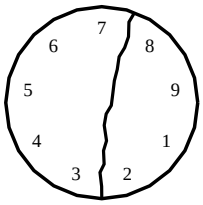
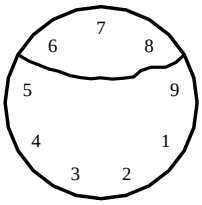
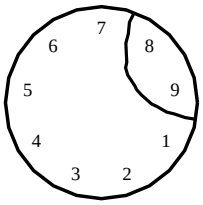
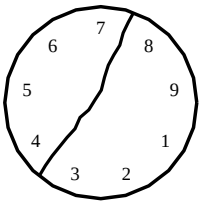
5. The balance is level: 4 melons and 1 orange on one plate, 2 melons and 7 oranges on the other.



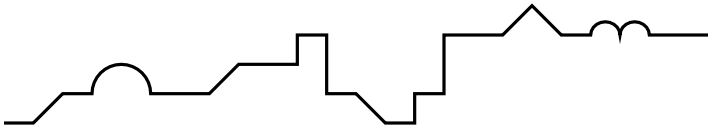
One melon weighs the same as how many oranges?

A. 7 oranges ***B.*** 6 oranges ***C.*** 4 oranges ***D.*** 3 oranges ***E.*** 2 oranges

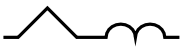
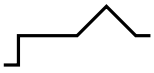
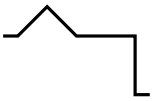
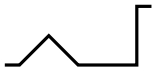
6. A round cookie with the numbers 1 to 9 around its edge is cracked into 2 pieces. The sums of the numbers in the pieces are two consecutive numbers.
Which picture shows how it is cracked?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

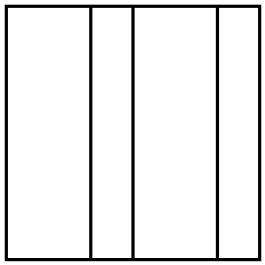
7. This is the outline of a town seen from the river at sunset.



Which piece **can** be part of the outline?

A.  ***B.***  ***C.***  ***D.*** ***E.*** 

8. A square garden is surrounded by a fence 32 m long. The gardener divides it into 4 rectangular beds as shown and puts a separate fence around each bed.



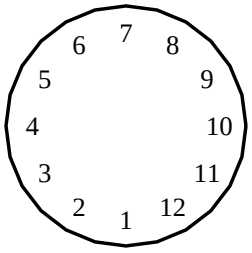
How many metres of fence are around the beds altogether?

A. 128 m ***B.*** 81 m ***C.*** 80 m ***D.*** 64 m ***E.*** 56 m

9. Hana, Aisha, Chloe, Leo and Zoe were born in different months: June, March, April, December and October. Aisha was not born in March. Leo was born in autumn. Hana was born in a month that starts with J. Chloe was born in winter.
Which statement is true?

A. Aisha was born in April ***B.*** Zoe was born in December ***C.*** Aisha was born in December
D. Hana was born in April ***E.*** Zoe was born in June

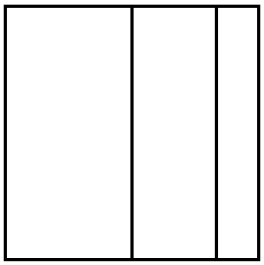
10. A round cookie with the numbers 1 to 12 around its edge is cracked into 3 pieces. The sums of the numbers in the pieces are three consecutive numbers.



What is the smallest of these sums?

A. 19 ***B.*** 24 ***C.*** 25 ***D.*** 26 ***E.*** 27

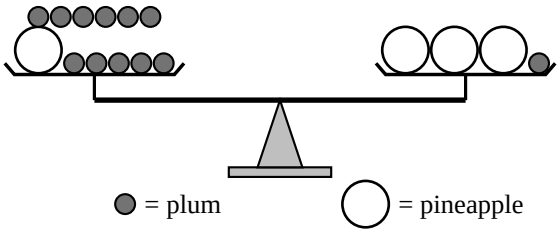
11. Grandma bakes a square cake. A ribbon 40 cm long fits exactly around it. She cuts the cake into 3 rectangular pieces as shown and puts a ribbon around every piece.



How many centimetres of ribbon does she need for all the pieces together?

A. 60 cm ***B.*** 70 cm ***C.*** 80 cm ***D.*** 82 cm ***E.*** 120 cm

12. The balance is level: 1 pineapple and 11 plums on one plate, 3 pineapples and 1 plum on the other.



One pineapple weighs the same as how many plums?

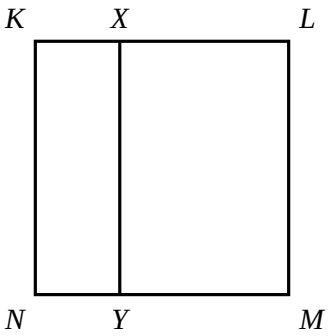
A. 12 plums ***B.*** 11 plums ***C.*** 10 plums ***D.*** 6 plums ***E.*** 5 plums

Part C: Each question is worth 5 points

13. Zoe, Ben, Tom, Emma and Omar each play a different instrument: the harp, the drum, the trumpet, the flute and the violin. Omar plays an instrument with strings. Zoe never uses sticks. Tom uses a bow. Emma blows into the instrument. Zoe doesn't play the flute.
Which statement is **not** true?

A. Ben plays the drum ***B.*** Emma plays the flute ***C.*** Emma plays the violin
D. Tom plays the violin ***E.*** Zoe plays the trumpet

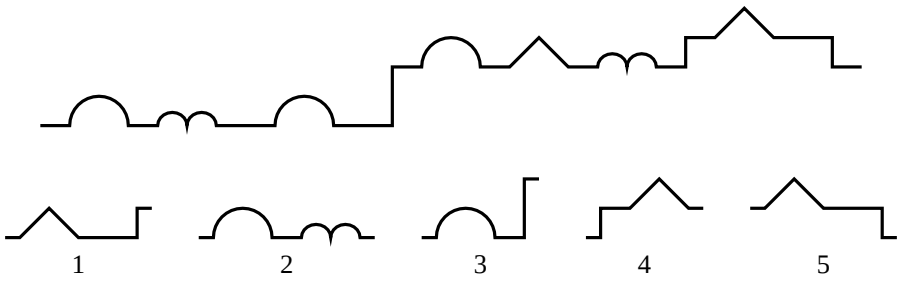
14. The square *KLMN* is cut into two rectangles. The perimeter of *KXYN* is 16 cm and the perimeter of *XLMY* is 20 cm.



How long is the side of the square *KLMN*?

A. 30 cm ***B.*** 24 cm ***C.*** 9 cm ***D.*** 6 cm ***E.*** 4 cm

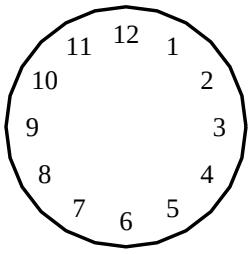
15. This is the outline of a town seen from the river at sunset.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

16. The face of a clock is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 3 and 4, and between 9 and 10

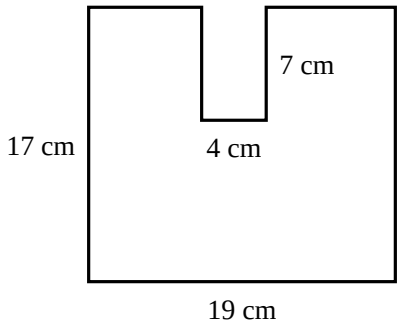
B. between 1 and 2, and between 10 and 11

C. between 4 and 5, and between 9 and 10

D. between 3 and 4, and between 7 and 8

E. between 11 and 12, and between 2 and 3

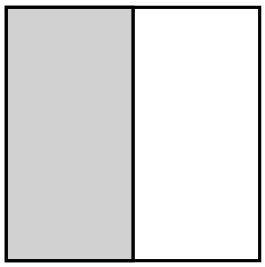
17. A 19 cm by 17 cm rectangle of cardboard has a rectangular notch 4 cm wide and 7 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 72 cm ***B.*** 79 cm ***C.*** 82 cm ***D.*** 86 cm ***E.*** 94 cm

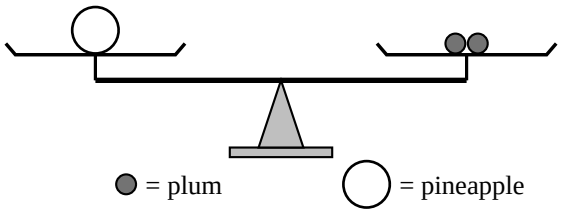
18. Mia cuts a square sheet of paper into 2 equal strips as shown. The perimeter of each strip is 42 cm.



How long is the side of the square?

A. 28 cm ***B.*** 14 cm ***C.*** 13 cm ***D.*** 11 cm ***E.*** 7 cm

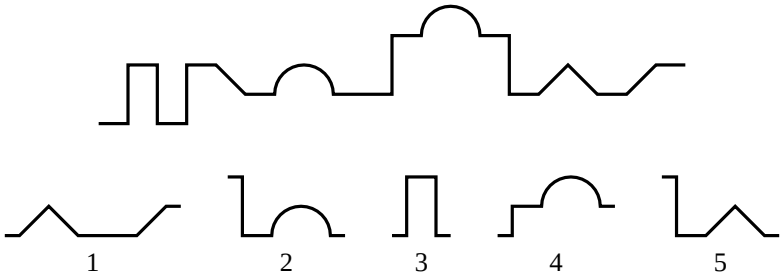
19. The balance shows that one pineapple weighs as much as 2 plums. 2 pineapples and 1 plum weigh 750 g altogether.



How much does one pineapple weigh?

A. 450 g ***B.*** 375 g ***C.*** 350 g ***D.*** 300 g ***E.*** 250 g

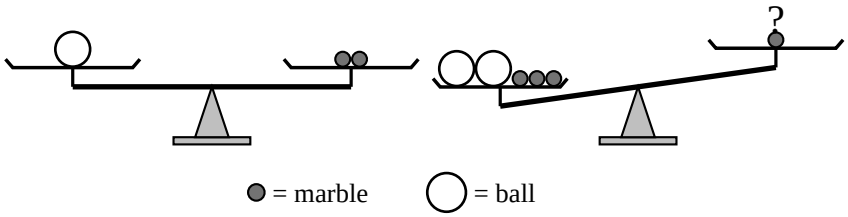
20. Far away we see the skyline of a castle.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

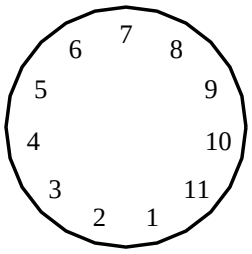
21. The first balance is level. On the second balance there are 2 balls and 3 marbles on the left and 1 marble on the right.



How many marbles must be added to the right plate to make it level?

A. 1 ***B.*** 6 ***C.*** 7 ***D.*** 8 ***E.*** 9

22. A round plate with the numbers 1 to 11 around its edge is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 7 and 8, and between 1 and 2

B. between 6 and 7, and between 1 and 2

C. between 7 and 8, and between 3 and 4

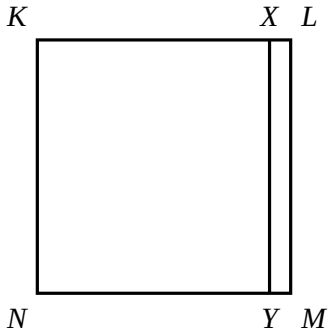
D. between 8 and 9, and between 2 and 3

E. between 10 and 11, and between 1 and 2

23. Fabienne, Emma, Leo, Nicolas and Ben each play a different instrument: the piano, the drum, the xylophone, the guitar and the harp. Emma plays an instrument with strings. Nicolas plays an instrument with strings. Ben plays with sticks. Emma doesn't play the harp. Leo doesn't play the xylophone. Leo plays with sticks.
Which statement is **not** true?

A. Fabienne plays the piano ***B.*** Nicolas plays the harp ***C.*** Emma plays the guitar
D. Fabienne plays the xylophone ***E.*** Leo plays the drum

24. The square *KLMN* is cut into two rectangles. The perimeter of *KXYN* is 46 cm and the perimeter of *XLMY* is 26 cm.



What is the perimeter of the square *KLMN*?

A. 48 cm ***B.*** 52 cm ***C.*** 53 cm ***D.*** 60 cm ***E.*** 72 cm

Answer Key

1.	**B**	**2.**	**B**	**3.**	**B**	**4.**	**D**	**5.**	**D**	**6.**	**E**
7.	**A**	**8.**	**C**	**9.**	**A**	**10.**	**C**	**11.**	**C**	**12.**	**E**
13.	**C**	**14.**	**D**	**15.**	**C**	**16.**	**A**	**17.**	**D**	**18.**	**B**
19.	**D**	**20.**	**B**	**21.**	**B**	**22.**	**D**	**23.**	**D**	**24.**	**A**

Solutions

1. B — All the other pieces can be found in the line, but this one cannot: it has a pointed roof where the line has two small bumps. [2002 Q7 · cannot_belong]

2. B (Julien) — The only possible matching: Omar – goldfish; Mia – hamster; Lucas – turtle; Julien – cat; Leo – dog. [2002 Q18 · who_has]

3. B (6 apples) — Take 2 pumpkins off both plates: 6 apples balance 3 – 2 = 1 pumpkin. So one pumpkin = 6 apples. [2002 Q9 · one_more_balances]

4. D (Emma) — The only possible matching: Emma – December; Julien – January; Tom – April; Mia – July; Noah – March. [2002 Q18 · who_has]

5. D (3 oranges) — Take 2 melons and 1 orange off both plates. Left: 2 melons = 6 oranges, so one melon = 6 ÷ 2 = 3. [2002 Q9 · both_sides]

6. E — All numbers add up to 45, so the pieces must be 22 + 23 = 45. Only this picture gives 22, 23. [2002 Q11 · which_picture]

7. A — Only this piece is found in the line. The others don't fit — B: its step is shorter than in the line; C: it is a mirror image of a part of the line; D: it has the right shapes but in the wrong order; E: it has the right shapes but in the wrong order. [2002 Q7 · can_belong]

8. C (80 m) — Side = 32 ÷ 4 = 8 m. There are 3 cuts; each cut is 8 m long and is counted twice (once for each piece next to it): 32 + 3 × 2 × 8 = 80 m. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

9. A (Aisha was born in April) — The only possible matching: Hana – June; Aisha – April; Chloe – December; Leo – October; Zoe – March. [2002 Q18 · which_true]

10. C (25) — All the numbers add up to 78. Three consecutive numbers that add up to 78: 25 + 26 + 27 = 78. The smallest is 25. [2002 Q11 · largest_sum]

11. C (80 cm) — Side = 40 ÷ 4 = 10 cm. There are 2 cuts; each cut is 10 cm long and is counted twice (once for each piece next to it): 40 + 2 × 2 × 10 = 80 cm. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

12. E (5 plums) — Take 1 pineapple and 1 plum off both plates. Left: 2 pineapples = 10 plums, so one pineapple = 10 ÷ 2 = 5. [2002 Q9 · both_sides]

13. C (Emma plays the violin) — The only possible matching: Zoe – trumpet; Ben – drum; Tom – violin; Emma – flute; Omar – harp. [2002 Q18 · which_not_true]

14. D (6 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 16 + 20 = 36 cm = 6 sides. Side = 36 ÷ 6 = 6 cm. [2002 Q6 · two_perimeters]

15. C (3) — Pieces 2, 4, 5 fit. Not: 1 (it steps up where the line steps down); 3 (its flat part between the two features has the wrong length). [2002 Q7 · how_many_fit]

16. A (between 3 and 4, and between 9 and 10) — Total 78, so each piece has 39: 4+5+6+7+8+9 = 39 and 10+11+12+1+2+3 = 39. [2002 Q11 · straight_crack]

17. D (86 cm) — The bottom of the notch (4 cm) replaces the 4 cm taken out of the top edge, and the two sides of the notch add 2 × 7 cm. Perimeter = 2 × (19 + 17) + 2 × 7 = 86 cm. [2002 Q6 · notch]

18. B (14 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 42 ÷ 6 = 7, so the side is 2 × 7 = 14 cm. [2002 Q6 · strips_to_square]

19. D (300 g) — Swap each pineapple for 2 plums: 2 × 2 + 1 = 5 plums weigh 750 g. One plum: 750 ÷ 5 = 150 g; one pineapple: 2 × 150 = 300 g. [2002 Q9 · grams]

20. B (2) — Pieces 3, 5 fit. Not: 1 (its flat part between the two features has the wrong length); 2 (it steps down where the line steps up); 4 (its step is shorter than in the line). [2002 Q7 · how_many_fit]

21. B (6) — One ball = 2 marbles, so the left plate is worth 2 × 2 + 3 = 7 marbles. The right plate has 1, so add 7 – 1 = 6. [2002 Q9 · what_to_add]

22. D (between 8 and 9, and between 2 and 3) — Total 66, so each piece has 33: 9+10+11+1+2 = 33 and 3+4+5+6+7+8 = 33. [2002 Q11 · straight_crack]

23. D (Fabienne plays the xylophone) — The only possible matching: Fabienne – piano; Emma – guitar; Leo – drum; Nicolas – harp; Ben – xylophone. [2002 Q18 · which_not_true]

24. A (48 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 46 + 26 = 72 cm = 6 sides. Side = 72 ÷ 6 = 12 cm, perimeter 4 × 12 = 48 cm. [2002 Q6 · two_perimeters]